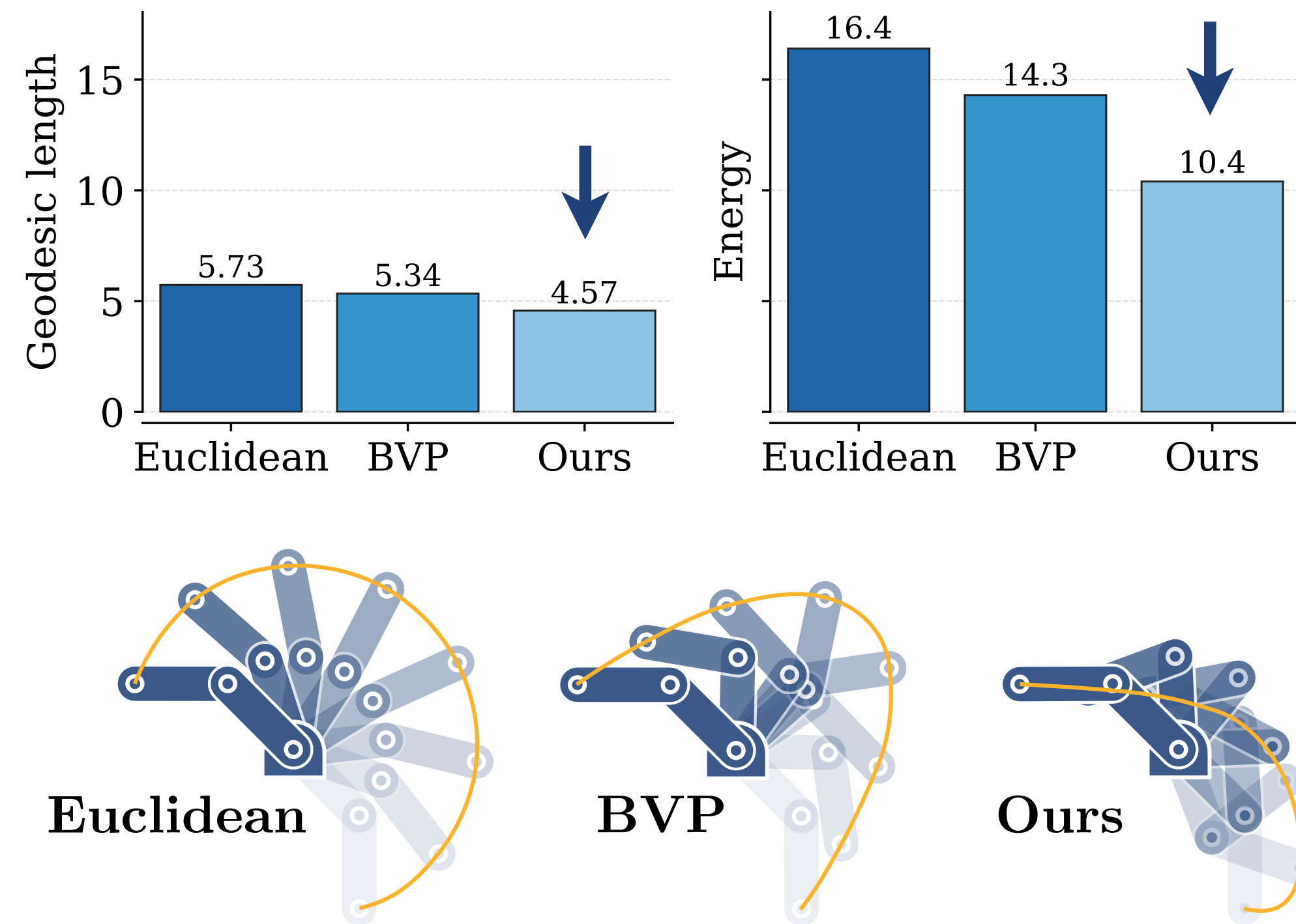
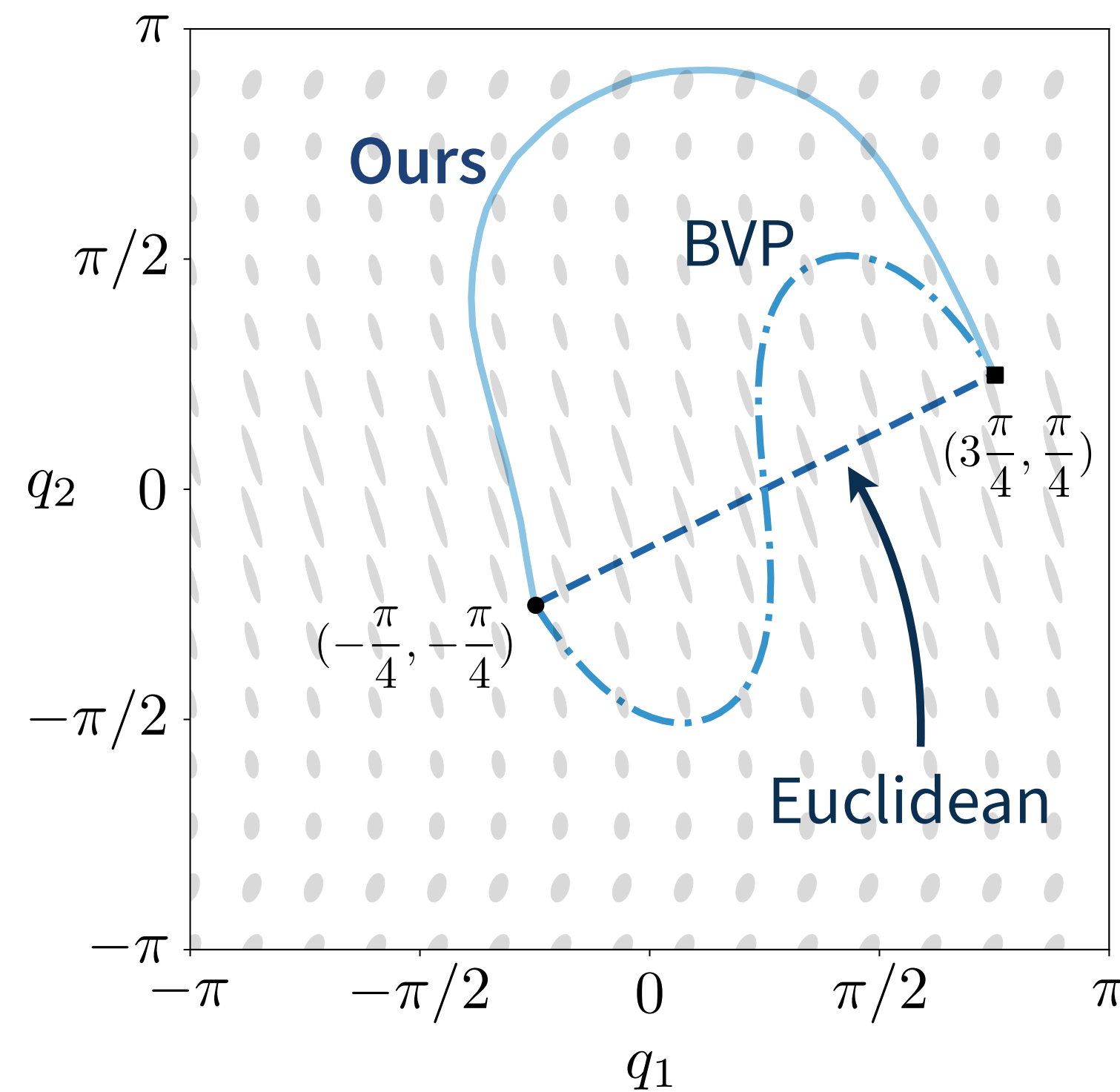


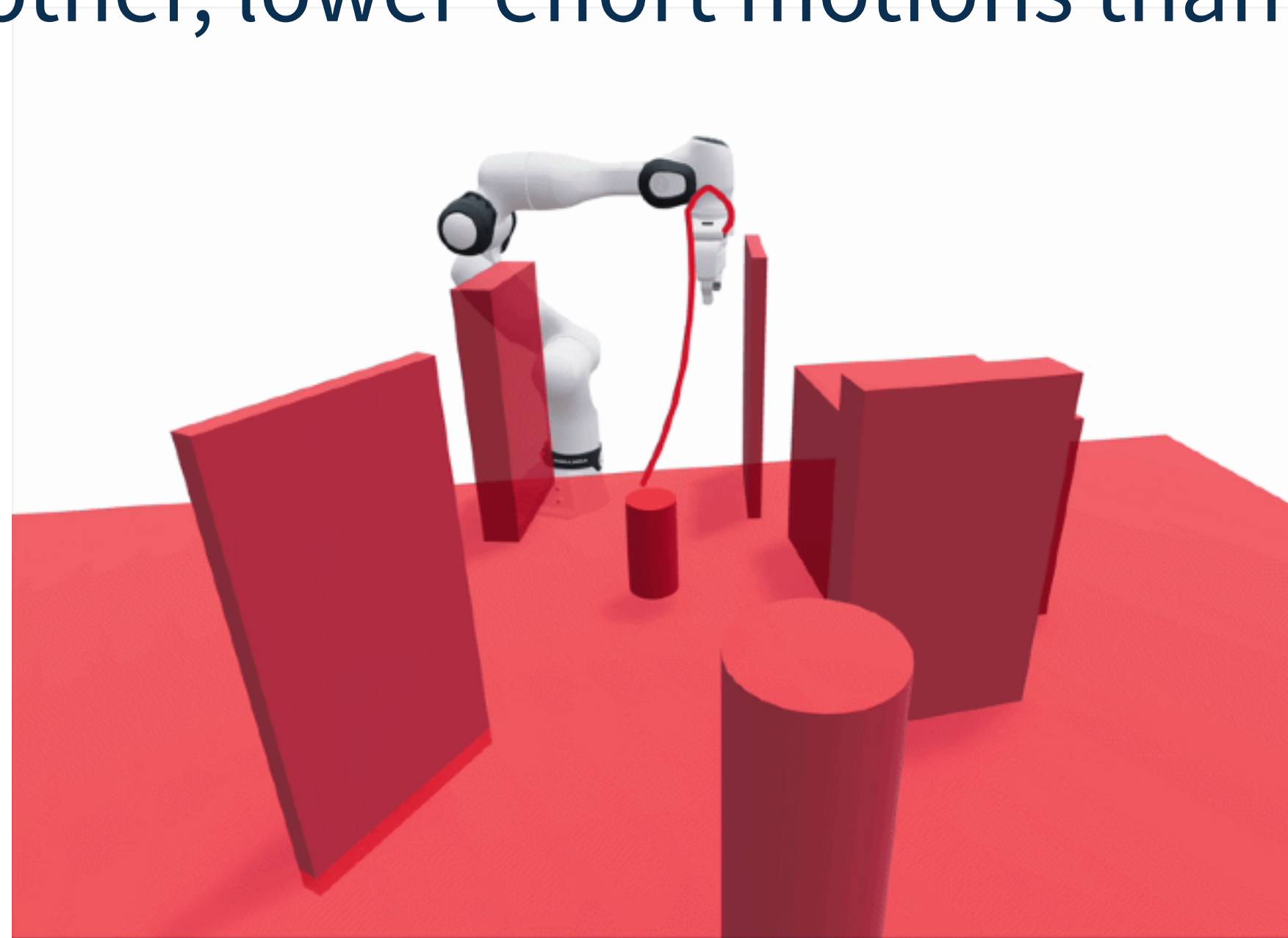
Shorter and lower-energy geodesics

— global search avoids the local-minimum trap



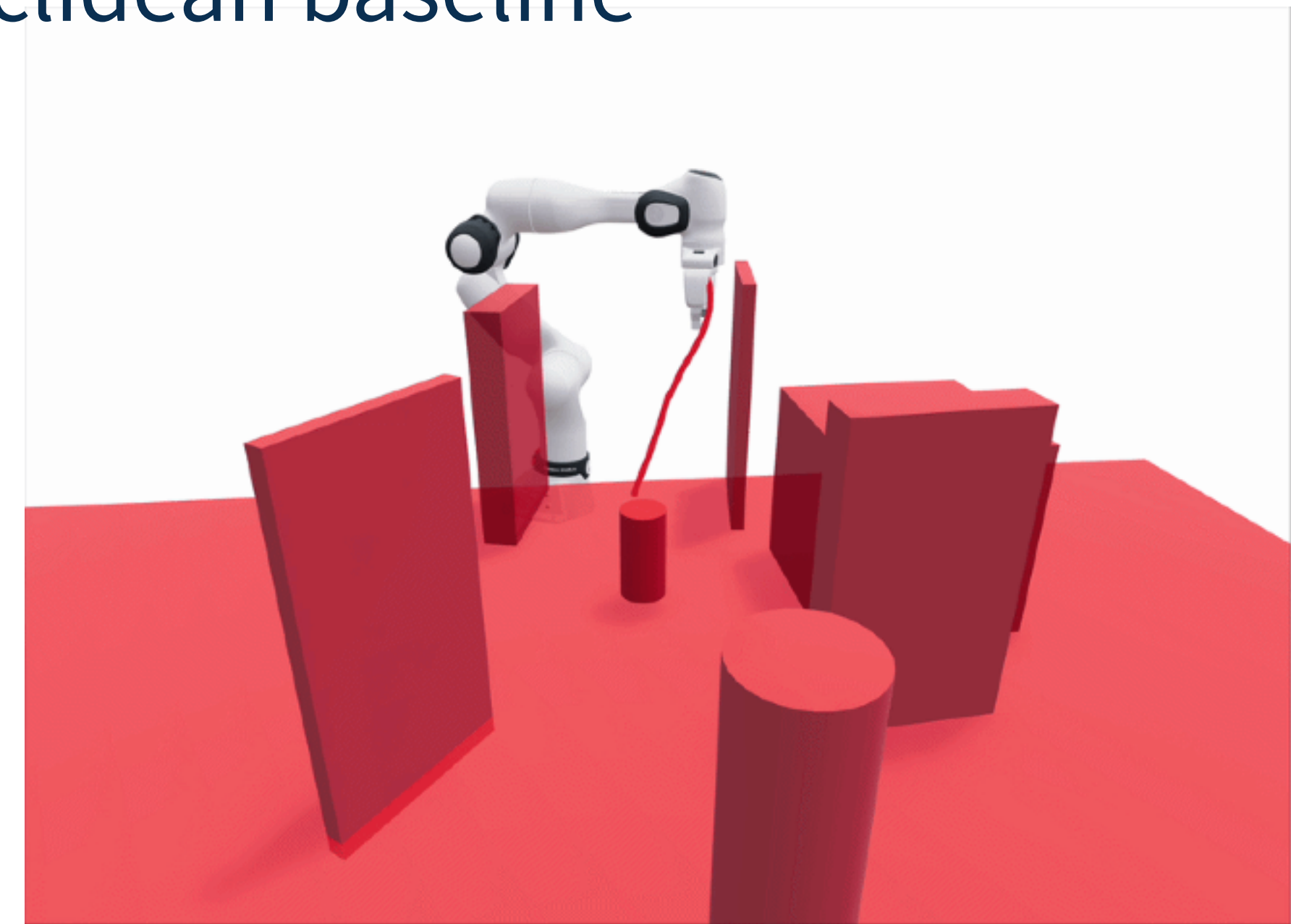
Numerical BVP solvers often need good initialization, else they tend to get stuck in local minima.

Scaling-up: the **kinetic-energy** metric produces smoother, lower-effort motions than the Euclidean baseline



Euclidean metric

joint-space straight-line motions



Kinetic-energy metric

joint-space motions that respect inertia

Optimization baselines often need hand-tuned barriers to avoid obstacles
— and can still get **stuck in local minima**.